AI-Driven Multi-Modal Adaptive Handover Control for O-RAN

Paper Summary and Thesis Positioning Notes

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1 Overview

Item	Content
Title	AI-Driven Multi-Modal Adaptive Handover Control Optimization for O-RAN
Author	Abdul Wadud, Fatemeh Golpayegani, Nima Afraz (University College Dublin)
Source	arXiv:2603.17158v1 · Adopted by IEEE ICC 2026 Workshop (WS-01)
field	O-RAN, handover, multi-modal mobility, Non-RT/Near-RT RIC, RL

One-line Summary. A hierarchical mobility-aware handover optimization structure in which the entire process of first classifying mobility modes (7 types), estimating future location and RSRP using mode-specific prediction models, and ranking handover candidate cells using PPO policies is placed in the **rApp** of a non-RT RIC, and the **xApp** of a near-RT RIC performs only standard-compliant handover decisions in a lightweight manner by combining that ranking with real-time E2 measurements.

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2 Motivation

The authors point out three limitations simultaneously.

(1) Mobility-agnostic problem of fixed A3 parameters. Traditional A3-event handover relies on fixed TTT / Hysteresis / CIO. Although simple and robust, a single fixed configuration is applied uniformly to all terminals and all mobile situations. Low-speed terminals experience ping-pong handovers caused by fading, while high-speed terminals experience late/failed handovers and RLF. Although optimal parameters should vary according to terminal speed and trajectory, this information is not used at all.

(2) Limitations of existing ML techniques that treat mobility only as a continuous variable. Actual mobility has a qualitatively distinct **multi-modal** structure such as pedestrian, bicycle, car, bus, and train, but existing ML handover treats this as a single continuous variable and fails to capture the characteristics of each mode.

(3) Computational burden of the near-RT layer. DQN/LSTM-based prediction techniques are accurate but heavy, making it difficult to handle long-term prediction contexts within the latency budget of the near-RT RIC.

Key Claim: There is no prior research that simultaneously satisfies multi-modal mobility awareness + predictive target ranking + lightweight real-time execution aligned with the O-RAN hierarchy.

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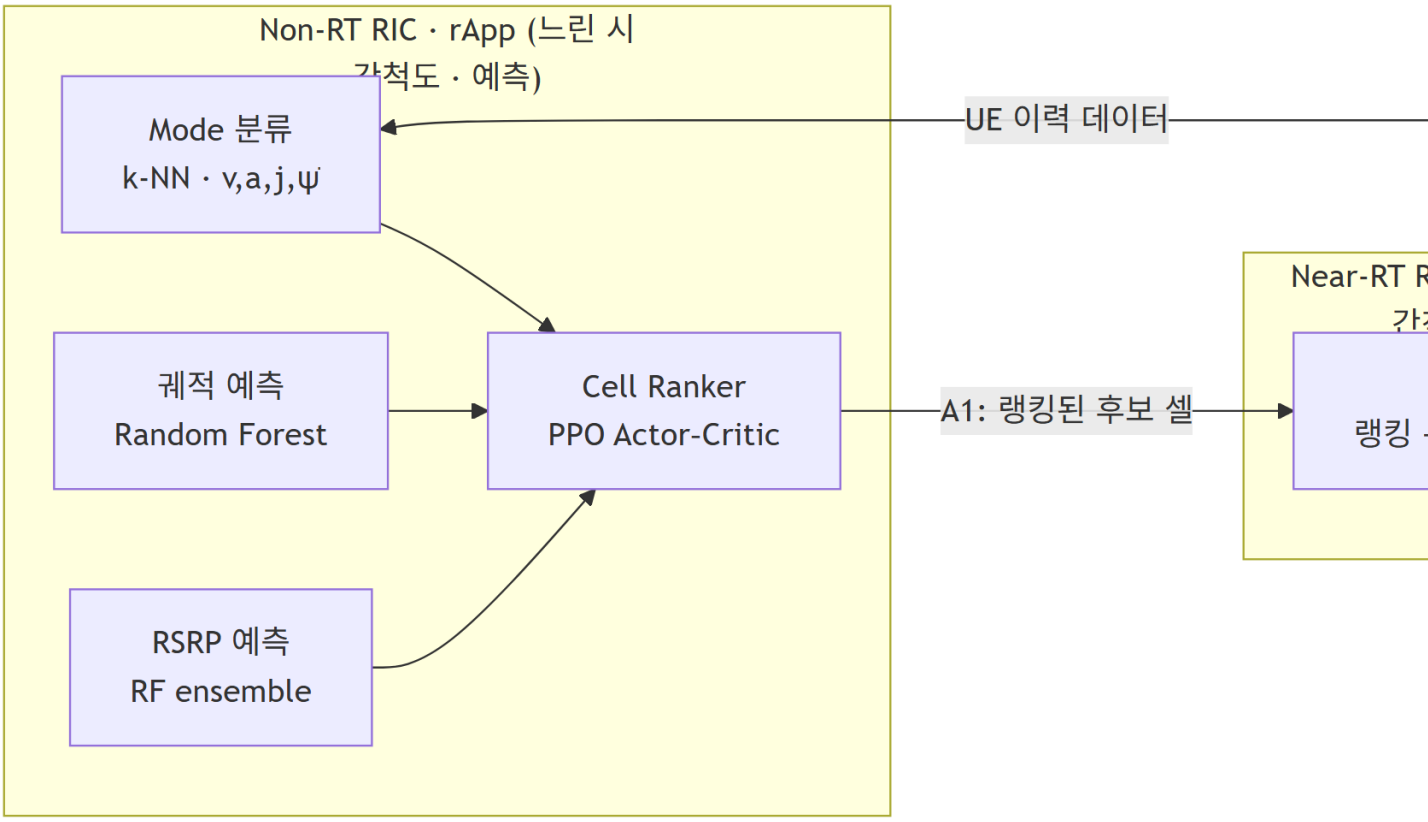
3 System Model & Architecture

The core design is the **layered separation of predictive intelligence (rApp, non-RT) and execution (xApp, near-RT)**, and the two are connected via the **A1** interface.



The design intent is clear. Heavy prediction and policy operations are handled by non-RT rApps with ample time, while leaving only light guard checks and standard-compliant execution to near-RT xApps to preserve low-latency execution. The fact that handovers are relatively slow decisions in sub-seconds enables this RIC-based externalization.

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4 Proposed Method Details

4.1 Mode Classification (MC)

A lightweight **k-NN** classifier receiving input $(v, a, j, \dot{\psi})$ + previous mode estimate. Here, jerk $j = d^3x/dt^3$ (rate of change in acceleration, trajectory smoothness) and bearing rate $\dot{\psi} = v\kappa$ (rate of change in direction of travel) are the key features distinguishing the modes. The output is the one-hot encoding $\mathbf{e}(m_t^u)$ of the 7 modes {ped, cyclist, car, bus, train, drone, uav}.

4.2 Trajectory Prediction (TP) · RSRP Prediction (P-RSRP)

Both are **Random Forest** regressions.

- **TP**: Next step position (v, a, m, x, y) $\hat{p}^u_{t+1}$ predicted
- **P-RSRP**: Cell-specific next-step RSRP vector $\hat{r}^u_{t+1}$ predicted by identical feature

The selection of a lightweight model instead of a heavy LSTM/DQN is the basis for the claim that "near-RT delay budget is not violated."

4.3 Cell Ranker (CR) — PPO Actor–Critic

- **state**: $s^u_t = [\,\mathbf{e}(m^u_t),\ \hat{p}^u_{t+1},\ \hat{r}^u_{t+1}\,]$ — Mode + Predicted Position + Predicted RSRP
- **action**: Select target candidate cell (ranking)
- **reward**: $r_t = \alpha\, w_u R \ - \ \beta\, H \ - \ \gamma\, PP$
  - R: throughput, H: handover penalty, PP: ping-pong penalty
- trained with clipped surrogate objective (PPO)

Only the ranking results are transmitted to the xApp via A1, and the xApp checks the load/QoS guards with live E2 measurements and then executes the final handover.

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5 Evaluation Results

5.1 Simulation Environment

MATLAB 5G Toolbox. O-CU 1 / O-DU 3 / O-RU 27, UE 100, approx. 7 km² urban area. 2.1 GHz, 20 MHz, transmit 46 dBm, UMa path loss, 8 dB shadowing. Fronthaul 10 Gbps (0.1–0.5 ms). Control interval 200 times (20 seconds) repeated 30 times, 95% CI.

Comparison group: Traditional A3 / O-RAN ML-Assisted (LSTM transfer learning) / Load Balance O-RAN.

5.2 KPI Comparison

indicators	MultiModal-AHC	Traditional A3	ML-Assisted	Load Balance
Throughput (Mbps)	77.10 ± 5.72	74.11	78.48	75.23
HO rate	0.0362	0.0962	0.0209	0.4013
Ping-pong (%)	**0.00**	0.00	12.64	11.29

5.3 Individual Model Performance

Module	Model	Metric
Mode Classification (MC)	k-NN	Accuracy 99.84%
Trajectory Prediction (TP)	Random Forest	RMSE 14.40 m, Correlation Coefficient 1.00
RSRP prediction (P-RSRP)	RF ensemble	RMSE 7.35 dB
Cell Ranker (CR)	PPO Actor–Critic	Average reward +23.4, HO failure approx. −18%, converges at 150–180 ep

The key message is the **stability–agility profile**. A3 has a ping-pong of 0 due to its conservative fixed TTT, but this technique achieves a ping-pong of 0 by predicting future position and RSRP and selecting only continuous and consistent targets. However, throughput is not statistically distinguishable as it overlaps with baseline and CI.

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6 Critical Evaluation

Weaknesses to note for future citation and differentiation.

The figures are excessively perfect. MC 99.84%, trajectory correlation coefficient 1.00, RSRP predicted mean −66.76 vs. actual −66.75. Since traces were *generated* with different kinematic models for each mode in MATLAB, it is highly likely that the classifier is separating data that was originally designed to be separated. The authors also left actual testbed verification as future work.

The deployment hierarchy of PPO is contradictory within the text. Abstract · Sec IV describes that PPO is in rApp(non-RT) and xApp makes the final decision, but Sec III-C refers to the same Actor–Critic as a "real-time decision engine." It is unclear whether the ranking is non-RT or real-time.

Terminology Confusion. It defines "HO rate = HO failure rate" and positively interprets the fact that the HO rate of MultiModal-AHC is *higher* than ML-Assisted as proactivity. The headline "18% failure reduction" is a separate figure in the PPO reward section rather than the KPI table, so the two metrics are mixed. Since there is no superiority in throughput, it is accurate to interpret the actual contribution as limited to **elimination of ping-pong + controlled HO rate**.

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7 Research Gap & Thesis Differentiation

This paper has two faces regarding the thesis.

7.1 Reusable Methodology Template

- **Direct correspondence of prediction building blocks**: The P-RSRP (channel quality prediction) and TP (traffic prediction) of this paper occupy the same positions as the channel quality prediction and traffic prediction required by the thesis.
- **Mapping to Heterogeneous Robot Environments**: $(v, a, j, \dot{\psi})$ The mobility mode classification based on this thesis is naturally mapped to the heterogeneous mobility environments (fixed robot arm / low-speed AMR / high-speed drone). This corresponds to the robot versions of the ped / car / drone modes in this paper.
- **predict-then-decide framework**: The structure that predicts future states and then proactively decides is the same framework as **Stage 4 (chance-constrained future PRB preassignment)** of the thesis blueprint.

7.2 Value as an architectural comparator (foil)

This paper places all intelligence in the non-RT rApp + near-RT xApp and remains within a **pure RIC paradigm** that operates solely with A1/E2/O1. This is possible because handover is a slow decision occurring in sub-second units.

This implicitly proves the core boundary of the thesis.

Control target	Time scale	Appropriate hierarchy	Rationale
Handover / mobility	sub-second	RIC (rApp/xApp, A1/E2)	This paper demonstrates
Importance-aware MAC scheduling	sub-TTI (hundreds of µs)	node collocation	cannot be externalized due to E2 loop latency

importance-aware MAC scheduling is sub-TTI control, so externalization of xApps via E2 is structurally impossible and GPU-collocated RAN+AI(pyAerial + GH200 + cuMAC) is required. Since this paper is a case where "predictive mobility control was successfully externalized in RIC," it serves as a stepping stone for positioning that **cites and differentiates at the same time**.

7.3 Differentiation Sentence (Draft)

> Predictive mobility control has been successfully externalized at the RIC layer (e.g., MultiModal-AHC, rApp→A1→xApp). However, the same externalization strategy cannot be applied to sub-TTI importance-aware MAC scheduling — because E2-based near-RT control loops (10 ms) do not cover decision cycles in the TTI range (hundreds of µs). This study is located at this point, namely the sub-TTI control loop that can be closed only at GPU-collocated RAN+AI nodes.

7.4 AI-RAN Frame Matching

The novelty of this thesis (Stages 3 and 4) lies in reducing the predict-then-allocate structure of this paper to sub-TTI, PRB, and node collocation units. In terms of tools, after algorithm prototyping, it leads to a verification path involving cuMAC validation on GH200 + pyAerial (actual CU/MAC/L1 access) and robot traffic generation using ROS2.

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8 Key take-away

1. This paper demonstrates that RIC externalization is valid for sub-second decisions, such as handovers.
2. The predict-then-decide skeleton for channel/movement prediction + mode classification + RL ranking is reusable.
3. Thesis differentiates itself by taking this framework down to the sub-TTI · PRB unit and placing it in a control loop that cannot be closed by RIC.